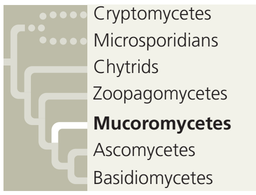
basidiomycetes; see Figure 31.10) have nonflagellated spores, which are dispersed by wind in terrestrial fungi.

Mucoromycetes



There are approximately 750 known species of **mucoromycetes**, fungi in the phylum Mucoromycota. This phylum includes species of fast-growing molds responsible for causing foods such as bread, peaches, strawberries, and sweet potatoes to rot during storage. Although some mucoromycetes are decomposers, most are associated with plants. Many mucoromycetes live as parasites or pathogens of plants, while others live as mutualists (including some mycorrhizae).

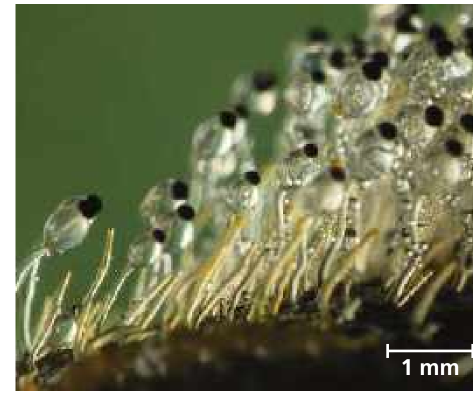
The life cycle of *Rhizopus stolonifer* (black bread mold) is fairly typical of mucoromycete species **(Figure 31.15)**. Its hyphae spread out over the food surface, penetrate it, and absorb nutrients. The hyphae are coenocytic, with septa found only where reproductive cells are formed. In the asexual phase, bulbous black sporangia develop at the tips of upright hyphae. Within each sporangium, hundreds of genetically identical haploid spores develop and are dispersed through the air. Spores that happen to land on moist food germinate, growing into new mycelia.

If environmental conditions deteriorate—for instance, if the mold consumes all its food—*Rhizopus* may reproduce sexually. The parents in a sexual union are mycelia of different mating types, which possess different chemical markers but may appear identical. Plasmogamy produces a sturdy structure called a **zygosporangium** (plural, *zygosporangia*), in which karyogamy and then meiosis occur. Note that while a zygosporangium represents the zygote ($2n$) stage in the life cycle, it is not a zygote in the usual sense (that is, a cell with one diploid nucleus). Rather, a zygosporangium is a multinucleate structure, first heterokaryotic with many haploid nuclei from the two parents, then with many diploid nuclei after karyogamy.

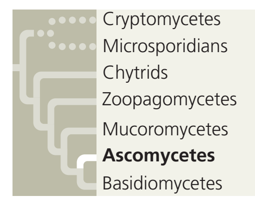
Zygosporangia are resistant to freezing and drying and are metabolically inactive. When conditions improve, the nuclei of the zygosporangium undergo meiosis, the zygosporangium germinates into a sporangium, and the sporangium releases genetically diverse haploid spores that may colonize a new substrate. Some mucoromycetes can actually "aim" and then shoot their sporangia toward bright light. **Figure 31.16** shows one example, *Pilobolus*, which decomposes animal dung. Its sporebearing hyphae bend toward light, where there are likely to be openings in the vegetation through which spores may reach fresh grass. The fungus then launches its sporangia in a jet of water that can travel up to 2.5 m. Grazing animals ingest the fungi with the grass and then scatter the spores in feces, thereby enabling the next generation of fungi to grow.

Finally, the phylum Mucoromycota also includes the glomeromycetes, a clade of fungi that form arbuscular mycorrhizae (see Figure 31.4b and Figure 37.14). The tips of the hyphae that push into plant root cells branch into tiny treelike arbuscules. About 85% of all plant species have mutualistic partnerships with arbuscular mycorrhizae.

▶ **Figure 31.16** ***Pilobolus*** **aiming its sporangia.**



Ascomycetes



Mycologists have described 90,000 species of **ascomycetes**, fungi in the phylum Ascomycota, from a wide variety of marine, freshwater, and terrestrial habitats. The defining feature of ascomycetes is the production of spores (called ascospores) in saclike **asci** (singular, *ascus*); thus, they are commonly called *sac fungi*. During their sexual stage, most ascomycetes develop fruiting bodies, called **ascocarps**, which range in size from microscopic to macroscopic **(Figure 31.17)**. The ascocarps contain the spore-forming asci.

Ascomycetes vary in size and complexity from unicellular yeasts to elaborate cup fungi and morels (see Figure 31.17). They include some of the most devastating plant pathogens, which we will discuss later. However, many ascomycetes are important decomposers, particularly of plant material. More than 25% of all ascomycete species live with green algae or cyanobacteria in beneficial symbiotic associations called lichens. Some ascomycetes form mycorrhizae with plants.

▼ **Figure 31.17 Ascomycetes (sac fungi).**

▶ *Tuber melanosporum* is a truffle species that forms ectomycorrhizae with trees. The ascocarp grows underground and emits a strong odor. These ascocarps have been dug up and the middle one sliced open.

◀ The edible ascocarp of *Morchella esculenta*, the tasty morel, is often found under trees in orchards.

? *Ascomycetes vary greatly in morphology (see also Figure 31.10). How could you confirm that a fungus is an ascomycete?*